Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/30/2013
PI Name:	Leong, Monica B.S.		
Project Title:	Packaged Food Mass Reduction Technology Development		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-3632
Organization Name:	Lockheed Martin/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2012	**End Date:**	10/01/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/13 per G. Douglas/JSC (Ed., 8/5/13)		
Key Personnel Changes/Previous PI:	NOTE: PI change in December 2012 from Maya Cooper to Monica Leong; see Cooper task with same title for prior reports (Ed., 2/25/2013)		
COI Name (Institution):	Cooper, Maya (NASA JSC/Lockheed Martin)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	The Advanced Food Technology (AFT) project seeks to define a safe, palatable, and nutritious food system to support long duration space missions beyond Low Earth Orbit (LEO). Balancing resources such as water, food preparation time, stowage volume, launch mass, and power requirements is central to defining a functioning food system solution for these missions. Prior AFT work has addressed the need to decrease the mass of the food system, and has recommended means such as reducing the moisture of the food system, increasing the nutrient density of the food system, and streamlining food packaging systems to accomplish this. Building on these recommendations, the objective of this project is therefore to identify technologies required to formulate meal replacement options that would decrease the mass and volume of the food system, as well as decrease the crew time associated with meal preparation. As commercially available meal replacement products are not adequate for NASA needs, this project will proceed by		

Task Description:	formulating new products that meet the unique requirements of spaceflight beyond LEO. The development of meal replacement options for the packaged space food system will be achieved by: * Aligning previous studies on nutrition and mass savings to compile the nutritional requirements and consumption guidelines for meal replacement beverages and bars. * Developing meal replacement bar(s) meeting nutrition, sensory, and shelf life requirements for space food systems. * Developing meal replacement beverage(s) meeting nutrition, sensory, and shelf life requirements for space food systems. * Assessing the meal replacement system and creating an implementation plan for the meal replacement options. Some accommodations have been made in the product development design to ensure that the project is completed within budget and on schedule. The recipe development of the bars and beverages will baseline from existing commercial bars and beverages rather than starting from new formulas even though significant changes are likely required to meet nutritional needs. Researchers will leverage flavor concepts from existing commercial bar and beverage products. Accelerated shelf life testing will done to assess stability as the project duration is shorter than the desired shelf life. While the ideal sensory evaluation would be a repeated exposure test to ensure acceptability levels were sustained over multiple experiences, the product amounts and the availability of panelists requires a one-time sensory evaluation for each product. Upon the completion of this study, an assessment of representative formulated meal replacement options will be provided to stakeholders, in order to offer an understanding of meal replacement technology and associated mass savings.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	The meal replacement technology developed during this task has applicability to the military and disaster relief agencies -- two organizations that use heavy amounts of shelf stable goods and are required to transport the goods over large distances. The development of nutritious meal replacement bars potentially alleviates significant dollars in logistical expenses by providing a lower mass, lower volume pathway to supplying adequate nutrition to the desired populations.
Task Progress:	The task work began by meal replacement bar and beverage prototypes being developed to meet macronutrient requirements for breakfast and lunch meals, based on current consumption guidelines. Additional food categories were identified as having low nutrient density and commercial food products and new formulations were proposed as replacements. Accelerated shelf life testing was conducted for all products to determine acceptability and general stability. Bars were confirmed to have a 48 to 72 week shelf life while beverages and most category replacements remained stable for 96 weeks. A variety of menus implementing these substitutions in the International Space Station (ISS) Standard Menu were proposed and estimated mass savings ranged from 11%-25%, and packaging savings of 16%-18% when one meal was replaced each day.
Bibliography Type:	Description: (Last Updated: 02/16/2015)
Abstracts for Journals and Proceedings	Leong ML, Cooper MR. "Packaged Food Mass Reduction Technology." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013